

ESG AND GREENWASHING DETECTION IN CORPORATE DISCLOSURES USING NATURAL LANGUAGE PROCESSING

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Abstract - Environmental, Social, and Governance (ESG) reporting has become central to how modern corporations communicate their sustainability efforts to investors, regulators, and the public. As companies face mounting pressure to demonstrate responsible business practices, ESG disclosures have grown exponentially in recent years. However, this surge in sustainability reporting has been accompanied by a rise in greenwashing - the practice of making misleading, exaggerated, or unsubstantiated claims about environmental and social impact. Many companies use aspirational language and vague commitments that sound impressive but lack concrete metrics or verifiable evidence of actual progress.

This project develops an automated NLP pipeline to systematically analyze corporate sustainability disclosures and detect potential greenwashing at scale. We processed sustainability reports and 10-K annual filings from major corporations, applying a multi-stage analysis that includes sentence extraction, ESG pillar classification (Environmental, Social, Governance), sentiment analysis, extractive summarization, and a novel

measurable-versus-vague classifier designed to flag unsubstantiated claims.

Our results reveal significant patterns in how companies communicate ESG matters. Most ESG-related sentences exhibit neutral sentiment, reflecting the descriptive nature of corporate reporting. More concerning, approximately 25–30% of analyzed sentences were classified as vague or unverifiable - relying heavily on phrases like "we are committed to," "we aim to achieve," or "we strive to improve" without accompanying quantitative data. This commitment-heavy but metric-poor language serves as a reliable indicator of potential greenwashing. The extractive summarization component successfully distills lengthy reports into coherent overviews, making it easier to identify discrepancies between stated goals and reported outcomes. Our work demonstrates that NLP techniques can effectively audit corporate sustainability claims at scale, providing a practical framework for improving transparency and accountability in ESG reporting.

Keywords— ESG, NLP, sustainability reporting, greenwashing detection, sentiment analysis, text mining

I. INTRODUCTION

Sustainability reporting has become essential to how companies communicate their environmental and social impact. Investors, regulators, and consumers increasingly rely on these reports to assess responsible business practices. However, as ESG disclosures expand, so does concern about greenwashing - when companies exaggerate sustainability performance or make vague commitments without evidence.

Recent studies show that many ESG disclosures contain phrases like "we are committed to" or "we strive to improve", which often appear without accompanying quantitative data. Detecting such greenwashing manually is resource-intensive and subjective, requiring analysts to review hundreds of pages looking for patterns that may indicate misleading claims.

Advances in Natural Language Processing provide an opportunity to automate this evaluation and bring transparency in sustainability communications. NLP techniques can analyze large volumes of text, identify ESG topics, assess sentiment, and flag potentially misleading language at scale.

This project investigates whether NLP can reliably extract ESG topics, evaluate sentiment, summarize disclosures, and detect greenwashed statements in corporate reports. Using sustainability reports and 10-K filings, we construct a pipeline that analyzes ESG language and identifies measurable versus vague statements, developing an interpretable framework for ESG transparency evaluation.

II. LITERATURE REVIEW

The Early work on domain-specific NLP for finance demonstrated quite clearly that adapting pretrained language models to financial text can greatly improve them. FinBERT is a BERT model that was fine-tuned on financial communications; it was introduced by Huang et al. (2023), and it outperforms generic models when doing sentiment analysis in financial documents. FinBERT and also similar models provide for a strong base for sentiment analysis within corporate sustainability disclosures, given that ESG reports often use finance-oriented language.

Gutierrez-Bustamante and others (2022) suggested a debut framework for scoring sustainability reports via NLP methods for sustainability reporting. In order to compare

company reports with the Global Reporting Initiative (GRI) standards, their study tested semantic similarity techniques like GloVe and Latent Semantic Analysis. This work shows converting narrative ESG reports into structured, measurable scores is feasible, though their approach used relatively shallow semantic models compared to transformer-based architectures.

Bronzini et al. (2024) recently examined how structured ESG views are derived from company releases. They did so via assigning pillar-level scores as well as handling cases where companies did not disclose relevant data. According to their work, automated methods can account for missing or inconsistent information. This is indeed relevant for this project because the disclosure quality varies quite widely across firms.

ESG-Miner, a system that automatically classifies news headlines into Environmental, Social, and Governance categories plus assigns sentiment-based ESG ratings, was developed beyond corporate reports by Fischbach et al. (2022). Their approach shows ESG performance can be approximated using even short media snippets of unstructured text. Potential extensions exist in my project thus sustainability reports can be cross-validated using external media sentiment.

Finally, Birti, Osborne, and Maurino (2025) put forth a proposal in order to optimize large language models for ESG activity detection. A new benchmark dataset called ESG-Activities was used by them. The dataset has more than 1,300 labeled text segments that map to EU ESG taxonomy categories, and the authors show synthetic data augmentation increases model performance. Their work shows just how one can fine-tune models with respect to ESG-related classification tasks. Also, their work provides a benchmark for evaluation.

Collectively, those studies suggest that though NLP analyses within ESG are still nascent, semantic similarity techniques steadily move to advanced transformer-based techniques. This project seeks to address gaps remaining within pillar-level classification and automated scoring. Also, it will be addressing the detection of vague as opposed to measurable claims.

III. METHODOLOGY

Our methodology implements a complete NLP pipeline for analyzing corporate ESG disclosures. The system processes raw PDF documents through multiple analytical stages, ultimately producing

structured outputs that quantify ESG content, sentiment patterns, and potential greenwashing indicators.

1. Data Collection

The analysis relies on two primary document types that represent different aspects of corporate disclosure:

- **Sustainability Reports** - These dedicated ESG documents typically range from 50 to 150 pages and contain details about environmental initiatives, social programs, and governance practices. Companies publish these reports voluntarily.
- **10-K Annual Filings** - These mandatory SEC filings contain audited financial information along with risk factors, governance structures, and material ESG-related disclosures. Unlike sustainability reports, 10-Ks are legally binding documents where companies must disclose material risks.

Documents were collected from publicly available sources and uploaded to the system in PDF format.

2. Text Preprocessing

The pipeline implements the following steps:

- **PDF Text Extraction** - We used PyPDF2 library to extract text from PDF files.
- **Sentence Segmentation** - The extracted text is processed through spaCy's sentence tokenizer. This step is critical because our analysis operates at the sentence level rather than document or paragraph level, making it easier to identify specific greenwashing claims.
- **Text Cleaning and Normalization** - We remove excessive whitespace, normalize unicode characters, and strip out non-semantic elements like bullet points and numbering artifacts.
- **Language Filtering** - We filter PDF's using language detection to ensure the models receive clean English input.
- **Quality Filtering** - Sentences shorter than 10 characters or containing fewer than 3 words are removed. Similarly, sentences

consisting entirely of numbers or special characters are filtered out.

3. ESG Pillar Classification

The first analytical step categorizes each sentence into one of three ESG pillars:

- **Environmental (E)** - climate change, emissions, energy use, waste management, water consumption, etc.
- **Social (S)** - labor practices, diversity and inclusion, employee health and safety, human rights, etc.
- **Governance (G)** - board composition, business ethics, regulatory compliance, risk management, etc.

We implemented a transformer-based sequence classification model fine-tuned on labeled ESG text. The model architecture uses a pre-trained BERT variant that has been adapted for ESG domain language. During inference, each sentence is tokenized and passed through the model, which produces probability scores for all three pillars. The pillar with the highest probability becomes the sentence's primary label.

The classification model outputs several useful metrics. At the sentence level, we capture the predicted pillar label along with confidence scores for each category. At the corpus level, we calculate pillar distributions showing what percentage of a company's ESG disclosure focuses on each area.

Additionally, we compute average sentiment scores per pillar, allowing us to identify whether companies discuss certain pillars more positively than others.

4. Sentiment Analysis

We implemented sentiment analysis using a FinBERT-based model specifically trained on financial text. The sentiment classifier assigns each sentence to one of three categories:

- **Positive** - achievements, favorable outcomes
- **Neutral** - factual statements, objective reporting

- Negative - risks, challenges, adverse impacts, concerns

Most ESG sentences fall into the neutral category, reflecting the descriptive nature of corporate reporting. However, the distribution of positive versus negative sentiment can indicate reporting bias.

5. Extractive Summarization

ESG reports often exceed 20,000 words, making manual review time-consuming. We used a transformer-based summarization model (DistilBART/BART-CNN) fine-tuned for long documents.

The system generates summaries at three levels: document-level overviews capturing the overall ESG narrative, pillar-specific summaries focusing exclusively on E, S, or G content, and section-level summaries for targeted analysis of specific initiatives. These condensed outputs help grasp key themes while enabling comparison between stated and actual measurements revealed through greenwashing detection.

6. Greenwashing Detection

The core innovation of this project is the greenwashing detection classifier, which distinguishes measurable claims from vague commitments. This binary classifier categorizes each ESG sentence as either:

- **Measurable** - statements backed by specific metrics. Examples include "reduced carbon emissions by 22% YOY" or "invested \$400 million in renewable energy projects."
- **Vague** – broad commitments, future-oriented goals without metrics, unverifiable claims.

The classifier was trained on a custom-labeled dataset of ESG sentences annotated for measurability. Training examples included both positive cases and negative cases.

The model identifies several linguistic patterns associated with vague claims:

- Future tense verbs
- Vague quantifiers
- Goals without timelines

Conversely, measurable claims typically include:

- Specific percentages or absolute numbers
- Year-over-year comparisons with data
- Completed past actions with outcomes

The greenwashing detector operates at sentence level, producing a binary label and confidence score. At the corpus level, we calculate the percentage of vague versus measurable sentences, providing an overall "greenwashing score" for each report. Higher percentages of vague language suggest potential greenwashing, particularly when combined with highly positive sentiment.

7. Output Aggregation and Reporting

The final stage consolidates results from all analytical components into structured outputs designed for both human interpretation and further computational analysis.

The system generates a comprehensive sentence-level dataframe containing:

- Original sentence text
- ESG pillar classification with confidence scores
- Sentiment label (positive/neutral/negative) with probabilities
- Greenwashing classification (measurable/vague)
- Document source and location metadata

At the document level, the system produces:

- Overall pillar distribution statistics
- Sentiment distribution per pillar
- Greenwashing percentage (ratio of vague to total sentences)
- Extractive summaries for each pillar
- Full document summary

All outputs are exported to a multi-tab Excel file for easy review:

- `sentence_output` tab contains the complete sentence-level analysis, allowing analysts to review individual claims and their classifications.
- `pillar_scores` tab provides aggregate statistics showing how much attention each ESG pillar receives and what sentiment characterizes discussion of each pillar.
- `pillar_summaries` tab includes the extractive summaries that capture key themes within each ESG dimension.

This structured output format supports multiple use cases. Investors can quickly assess whether a company backs its ESG claims with data. Regulators can identify companies making unsubstantiated commitments. Researchers can perform cross-company comparisons to benchmark reporting quality. The sentence-level granularity ensures that all findings can be traced back to specific source text, maintaining transparency and auditability throughout the analysis process.

IV. RESULT

1. Analysis and Findings

Distribution of the Data

The analysis yielded results for 35 corporate documents and demonstrated that there is a large difference in the total amount of ESG content in each type of document. Sustainability reports included significantly higher quantities of ESG information than 10-K filings. For example, there were 10,418 ESG-related sentences in the sustainability reports versus 4,167 sentences contained in the 10-K filings.

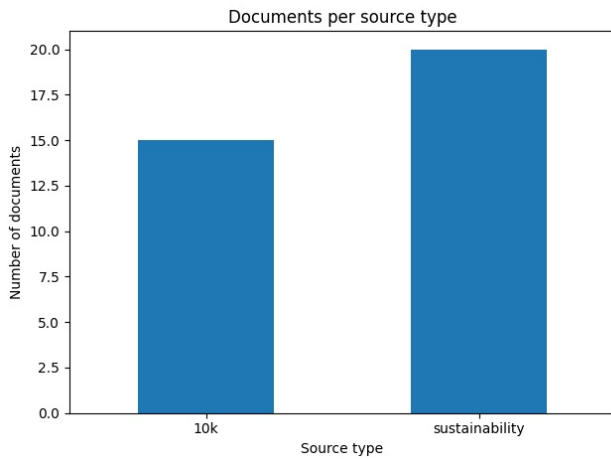


Fig. 1. Document count per source type.

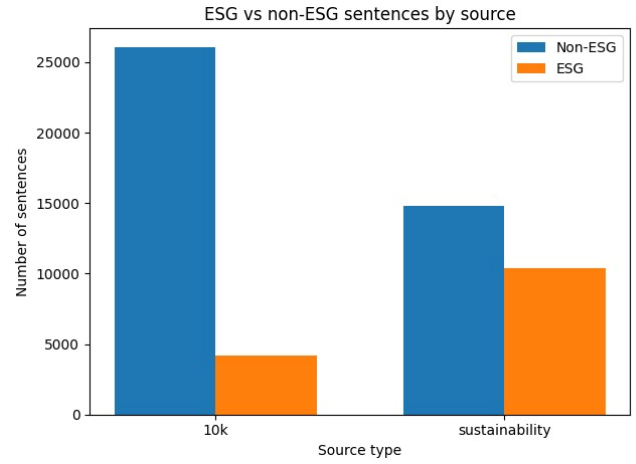


Fig. 2. Comparison of ESG vs. Non-ESG sentence distribution across document types.

Classification by Pillars

The multi-label classification of the documents resulted in a set of prioritized reporting formats: Sustainability Reports had the largest amount of reporting focus on the Environmental Pillar. Next most reporting focus was on the Social Pillar, with Governance Pillar having the least focus.

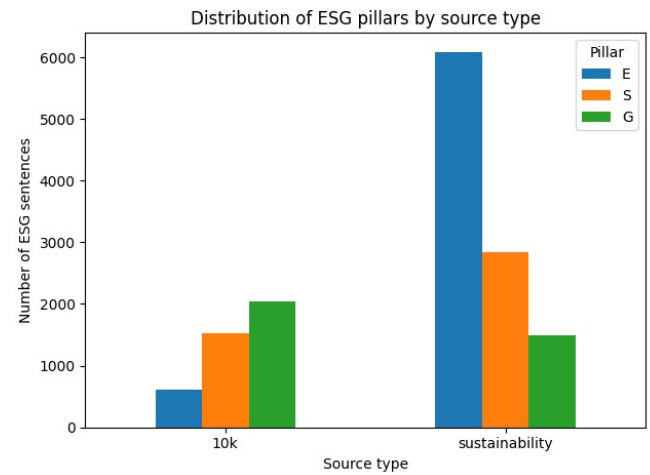


Fig. 3. Distribution of ESG pillars (Environmental, Social, Governance) by source type

In comparison, 10-K filings had their reporting priority structured in reverse from that of sustainability reports. The Governance Pillar was the top priority, followed by the Social Pillar, and lastly the Environmental Pillar. The analysis of the 10-K filings shows a strong emphasis on regulations concerning risk and compliance.

Detection of Greenwashing (Analysis of Vagueness)

For detection of vagueness, our classification of measured vs vague was utilized to provide the type of claims made by the reporting companies.

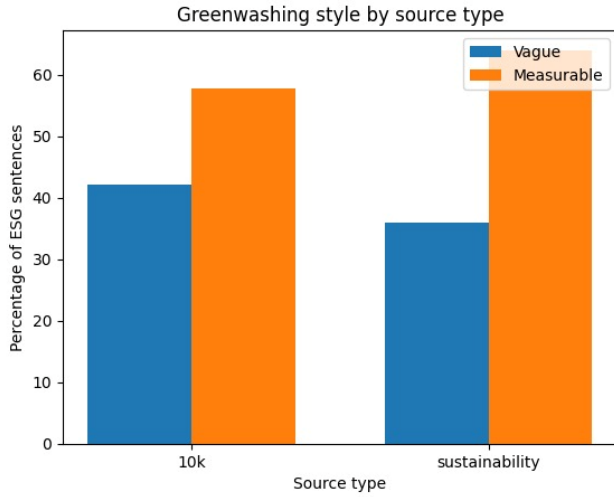


Fig. 3. Percentage of Vague vs. Measurable claims in 10-K filings and Sustainability reports

10-K Filings were ranked as having 42.2 % of ESG-related sentences classified as vague (1,757 / 4,167), while 57.8% were classified as measurable. **Sustainability Reports** had 36.0% of their sentences classified as vague (3,750 / 10,418) while 64.0% were classified as measurable.

Sentimental Analysis

Sentiment scores on the two document types exhibited neutral to slight positive sentiment. The sustainability reports were slightly more positive in sentiment (Average Sentiment Pillars of E and S have Values > 0.005) in comparison to the 10-K filings which were approximately neutral to slightly negative (with Average Sentiment Scores in 10-K filings = -0.003 and through 0.001) consistent with the factual and conservative tone of financial filings.

V. OUTCOMES

The main goal of this project is to create a complete and reliable natural language processing (NLP) system to help automate the often-tedious task of auditing sustainability reports. This system has produced three primary outcomes from a technical perspective:

Context-Aware Classification: We have built a specialized transformer model that accurately categorizes sustainability report text as either Environmental, Social or Governance. This allows for greater detail in analyzing report content at this level.

Quantifying the Vagueness Score: By creating a binary classifier to classify language as "measurable" or "vague," we provide a concrete metric for measuring greenwashing. This moves the discipline from relying on subjective interpretations to a standardized numerical score.

Automated Summarization: The Distil BART summarization tool allows for the compression of lengthy (>100 page) sustainability reports to concise pillar-based summaries that significantly lessen the amount of time and mental processing required to review lengthy reports.

VI. DISCUSSION

The results indicate the disparity between Corporate Disclosure frameworks and therefore illustrate to Stakeholders, that they cannot rely solely on one document to evaluate the ESG health of a company. There is evident task division; Sustainability reports tell the Environmental "story" and provide tabular data as support for their report, while 10-Ks primarily focus on Governance and risk. Interestingly, the legally mandated 10-Ks contained a higher quantity of ambiguous wording in comparison to the Sustainability Reports (which are voluntary). This indicates that under the threat of legal repercussions for regulatory filings, Companies tend to use defensive, non-committal phrasing, i.e., "we expect to", whereas because Sustainability Reports have a marketing-driven focus, Companies are encouraged to make tangible and quantifiable claims about their ESG Achievements.

While on the surface, it appears that voluntary reporting has a more solid foundation than mandatory reporting, the fact that over one-third of the statements/claims made in voluntary reports are still ambiguous, is a serious "red flag". A company that combines a high level of positive sentiment and has a large volume of promises for future performance, as evidenced by the use of phrases like "we intend to" or "we are striving to", without any supporting historical information is building a "commitment-heavy, metric-poor" frame of reference. Therefore, in order to conduct a complete ESG audit, one must take into consideration both the measurable claims from voluntary reports and the risk disclosures contained

in mandatory filings, as neither of those forms of documentation provide the whole story.

VII. CONCLUSION

This research paper demonstrates how Natural Language Processing will act as a reliable means of assessing corporations' compliance with Environmental, Social and Governance (ESG) criteria. The researchers move beyond traditional methods of evaluating the sentiment of corporate statements with Natural Language Processing and have instead looked at the nature of the claims companies are making, whether the claims can be measured or are ambiguous, in order to create an automatic means of determining the quality of ESG reporting.

This research shows that the common assumption that regulatory filings provide the most accurate ESG data may not always hold true, as many times, companies' voluntary disclosures contain more accurate and quantifiable metrics. As a result, Natural Language Processing allows for the identification of unsubstantiated corporate commitments very quickly and enables regulators and investors to focus their verification of companies where verification is most needed. Future efforts in research will expand on this work by comparing the ESG claims made by companies through their disclosures with the actual implementation of such claims as reported by other sources in the media.

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